



# Ali Haidar

**AI/ML & Robotics Engineer | Computer Vision · Deep Learning · Applied ML in Automation**

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## SUMMARY

AI/ML Engineer with an MSc (Honors) in Robotics & AI from ITMO University and hands-on delivery across research, industry, and freelance work — spanning applied generative AI, deep learning-based computer vision, and production ML. Built an agentic RAG chatbot backend (FastAPI, LangGraph, Weaviate) for bilingual restaurant and gym clients, and applied regression modeling and similarity-based matching for cost estimation at IROS-System. Extends this AI/ML foundation into physical systems through robotics work (motion control, kinematics, ROS), bringing research-grade models into real-world automation.

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## EXPERIENCE

### Freelance AI/ML Engineer — Self-employed

Aug 2025 – Present

*Remote*

- Designed and shipped an AI agent for a restaurant chain's RAG chatbot backend (bilingual RU/EN menu + FAQ), using FastAPI, LangGraph, and Weaviate Cloud for retrieval.
- Implemented logging, monitoring, and observability with Arize Phoenix to track and debug agent behavior in production.
- Delivered an AI agent-based chatbot engagement for a gym client.

### AI/ML & CAD Engineer — Industrial Automation — IROS-System

Jul 2023 – Present

*Saint Petersburg, Russia*

- Built a cost-prediction regression model and a cosine-similarity-based part-matching system (Python, pandas, scikit-learn) from BOM and assembly data, enabling earlier cost estimates and reducing part duplication.
- Designed 3D part models, assemblies, and technical drawings, and built rule-based automation scripts (iLogic, Inventor API) to auto-generate design variants and drawings, cutting modeling time and reducing manual errors.
- Performed stress calculations and FEA validation across all design outputs, maintaining 100% compliance with regulatory standards.

### Research Assistant & Robotics Engineer — Higher Institute for Applied Sciences and Technology (HIAST)

Oct 2018 – Oct 2022

*Damascus, Syria*

- Supervised 4th- and 5th-year capstone projects in robotics and mechatronics, including a Knee CPM machine, a propulsive bionic foot, and a laser engraving machine.
- Taught Industrial Drawing and CAD to 100+ students; led advanced modules in Robotics, Industrial Automation, and CAM.
- Delivered practical training in Ansys simulation and PLC coding, preparing students for industrial manufacturing roles.

### Mechatronics Engineer, Automated Systems (Part-time) — Syriatel Mobile Telecom and Milkman

Jun 2018 – Jun 2020

*Damascus, Syria*

- Designed and implemented a custom 500×500×500mm Cartesian 3D printer, managing the full lifecycle from mechanical design to firmware integration.
- Developed a high-precision volumetric filling machine, a cup feeder machine, and an end-line picker machine for high-viscosity, food-grade production lines.

- Managed development of a Building Management System (BMS), improving facility automation and energy efficiency.

## EDUCATION

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### **MSc (Honors), Robotics and Artificial Intelligence — ITMO University**

Oct 2023 – Jul 2025

*Saint Petersburg, Russia*

- Graduated with Honors (Red Diploma).
- Thesis: "Real-Time Visual Object Tracking Methods with Deep Learning Techniques."
- Coursework: Machine Learning in Robotics, Computer Vision, ROS, Motion Control, Path Planning, Cyber-Physical Systems, Big Data Processing.
- Research focus: implementing deep learning algorithms for visual perception in dynamic environments.

### **Russian Language Preparatory Program — ITMO University**

Oct 2022 – Sep 2023

*Saint Petersburg, Russia*

### **BEng, Mechatronics Engineering — Higher Institute for Applied Sciences and Technology (HIAST)**

Oct 2013 – Oct 2018

*Damascus, Syria*

- Graduated Top Performer (Ranked 'A' for distinguished academic performance); final grade 82.55%.
- Graduation project: designed and manufactured a Delta Linear Robot with 20μm positioning accuracy for laser drawing and 3D printing applications.
- 4th-year project: designed an open-wheel electric vehicle (Go-Kart).

## SKILLS

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**AI/ML & Data Science:** Machine Learning, Deep Learning, Computer Vision, Artificial Neural Networks, Image Processing, Reinforcement Learning, TensorFlow, Python (pandas, scikit-learn)

**Generative AI & LLMs (applied):** Retrieval-Augmented Generation (RAG), AI Agents / LangGraph, Large Language Models (LLMs), Generative AI

**Robotics:** Robot Operating System (ROS), Motion Control & Kinematics, Path Planning, Mechatronic System Design

**Engineering & CAD:** Autodesk Inventor, Autodesk Vault, AutoCAD, SOLIDWORKS, Computer-Aided Design (CAD), Finite Element Analysis (FEA), CAD Standards & Bill of Materials, Technical Drawing, CNC Manufacturing, 3D Printing, MATLAB

**Tools & Infrastructure:** FastAPI, Weaviate, Docker, Git, Programmable Logic Controller (PLC), iLogic

## CERTIFICATIONS

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- Generative AI with Large Language Models — DeepLearning.AI (Jul 2026)
- Retrieval Augmented Generation (RAG) — DeepLearning.AI (Jul 2026)
- AI Agents in LangGraph — DeepLearning.AI (Jun 2026)
- Functions, Tools and Agents with LangChain — DeepLearning.AI (Jun 2026)
- LangChain for LLM Application Development — DeepLearning.AI ([NEED DATE])
- LangChain Chat with Your Data — DeepLearning.AI ([NEED DATE])
- Knowledge Graphs for RAG — DeepLearning.AI (Jun 2026)
- Robotics and Artificial Intelligence (Sber × ITMO) — Sberbank-Technology (Jul 2025)
- Docker Certified Associate Exam Prep Specialization — LearnKartS (Jul 2026)
- Introduction to Git and GitHub — Google (Feb 2025)
- Introduction to CAD, CAM, and Practical CNC Machining — Autodesk (Mar 2021)

## LANGUAGES

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Arabic (Native) · English (B2) · Russian (B1)